

RICHARD SARTELL

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Education

University of Minnesota

Sep. 2022 – Expected Dec. 2026

Bachelor of Science: Electrical & Computer Engineering

Minneapolis, MN

- Embedded Sys. Design
- Analog Electronics I-III
- Signals and Systems
- DSA I-II
- Communication Systems
- Computer Architecture
- Intro. Microcontrollers
- Operating Systems

Experience

Boston Scientific

May 2026 – Present

R&D Systems Engineering Intern

Arden Hills, MN

- Enabled runtime streaming of medical imaging platform by designing and implementing Bash-based Linux Internet Connection Sharing service.
- Automated Linux Ethernet interface discovery, IP configuration, IPv4 forwarding, DNS forwarding, Network Address Translation, and firewall rule management using iptables and systemd.
- Developed automated Linux diagnostic service capturing network interface state, systemd service health, IP addressing and routing, IPv4 forwarding, and firewall/NAT configuration for maintenance verification.
- Improve development team validation, software deployment, and troubleshooting by verifying installation procedures and authoring technical documentation.

Punch Through

Feb. 2025 – Apr. 2026

Associate Embedded Software & Electrical Engineer

Minneapolis, MN

- Developed Python/Pytest HIL and E2E tests for embedded firmware; including BLE attenuation and PC CPU contention stress scenarios.
- Maintained and troubleshot Jenkins CI pipelines supporting embedded test automation.
- Delivered a one-revision KiCad PCB layout for client testing infrastructure.
- Contributed to 100% team firmware unit test coverage through VectorCast; performed MISRA-C compliance checks using IAR Embedded Workbench.

STRIVE Medical UMN

Sept. 2025 – May 2026

Lead Firmware Engineer

Minneapolis, MN

- Led firmware development for a Teensy 4.1-based robotic surgical simulation arm across multiple contributors.
- Coordinated firmware architecture, priorities, and interfaces across electrical, software, and project leadership teams.

Projects

Smart Network-Attached Storage | *Senior Design Project*

- Self-engineered firmware development on Raspberry Pi Pico 2 W to process captive-portal requests for user control of automated wireless file backup.
- Validated over-voltage protection circuit design through electrical stress testing, following structured stage-by-stage testing methods.
- Evaluated PCB design updates for over-voltage protection and connector integration.

Syndex | *STRIVE Medical*

- Develop Teensy 4.1 firmware for a robotic surgical simulation arm, supporting communication with Unity host via USB.
- Implemented embedded state-machine logic and firmware infrastructure for I2C encoder signal acquisition, custom USB packet communication, and device control.
- Defined firmware architecture and development workstreams across state management, encoder integration, and host-device protocol design.

Electrocardiogram System | *Junior Design Project*

- Designed and verified Altium PCB for ECG signal acquisition using differential arm heartbeat signals with leg-driven common-mode feedback path for signal noise cancellation.
- Conditioned output signals for digital conversion using an INA128P instrumentation amplifier and two AD708JN operational amplifiers, transmitted to a Python graphing script via PIC24FJ64's UART.

Skills

Programming and Scripting: C/C++, Python, Linux Bash, PowerShell

Version Control and CI/CD: Jenkins, Azure Pipelines, Git, GitHub, Bitbucket

Hardware and EDA Tools: KiCad, Altium Designer, DMMs, power supplies, oscilloscopes, arbitrary function generators

Testing and Development Tools: IAR Embedded Workbench, MPLabX IDE, VectorCast, Jira